

MapReduce Combiner Example with sensor data :

Let us explain a **classic Hadoop MapReduce optimization** using a **Combiner**.

The core topic is:

How to reduce work on Reducer and reduce network data transfer by doing local aggregation at Mapper side.

This is **very important** in Hadoop interviews and real-world Big Data systems.

1. Base Problem :

You have a **300 MB sensor data file**.

Each row contains:

Date	Time	Temperature
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12/12/2015	00:00	50
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12/12/2015	01:00	52
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12/12/2015	02:00	49
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12/13/2015	00:00	48
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12/13/2015	01:00	54
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12/13/2015	02:00	55
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Goal:

Find:

Maximum temperature for each day

Output:

Date	Max Temp
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12/12/2015	52
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12/13/2015	55
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2. HDFS Block Splitting

Default block size = **128 MB**

300 MB file becomes approximately:

- Block 1 = 128 MB
- Block 2 = 128 MB
- Block 3 = Remaining

So Hadoop creates:

- ✓ 3 Input Splits
 - ✓ 3 Mappers (usually)
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3. Without Combiner (Image 1)

Mapper Work

Each mapper reads records and emits:

(date, temp)

Example:

Mapper1:

(12/12,50)

(12/12,52)

Mapper2:

(12/12,49)

(12/13,48)

Mapper3:

(12/13,54)

(12/13,55)

4. Shuffle Phase

All mapper outputs go through network to reducer.

Reducer receives:

12/12 -> 50,52,49

12/13 -> 48,54,55

Then reducer calculates max:

12/12 -> 52

5. Problem Here

Mapper is doing almost no intelligent work.

Reducer is doing:

- grouping
- max logic
- handling many values
- network receives all raw data

This causes:

- ✗ More network traffic
 - ✗ Reducer overload
 - ✗ Slower job
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6. Combiner Concept :

Combiner is called:

Mini Reducer running after Mapper locally

Runs on same machine where mapper runs.

It performs **local aggregation before sending data to reducer.**

7. With Combiner

Same mapper outputs:

Mapper1:

(12/12,50)

(12/12,52)

Mapper2:

(12/12,49)

(12/13,48)

Mapper3:

(12/13,54)

(12/13,55)

Now combiner runs locally.

Combiner Output

Mapper1 Combiner:

12/12 -> 52

Mapper2 Combiner:

12/12 -> 49

12/13 -> 48

Mapper3 Combiner:

12/13 -> 55

8. Now Reducer Receives Less Data

Instead of 6 values, reducer gets only 4 values:

12/12 -> 52,49

12/13 -> 48,55

Reducer final output:

12/12 -> 52

12/13 -> 55

Same answer. Less work.

9. Why Combiner Helps

Advantage 1: Less Network Traffic

Before:

6 records moved

After:

4 records moved

In TB scale data, this is huge.

Advantage 2: Faster Reducer

Reducer gets partially processed data.

Advantage 3: Better Parallelism

Work shared among many mappers.

10. Real World Example

Suppose 1000 servers processing logs.

Without combiner:

All raw counts go to reducer.

With combiner:

Each server counts locally first.

Then sends summary only.

Huge speed boost.

11. Important Rule: Combiner is Optional

Hadoop may run combiner:

- 0 times
- 1 time
- multiple times

So never depend on combiner for correctness.

Use only for optimization.

12. When Safe to Use Combiner

Use when operation is:

Associative + Commutative

Means grouping order doesn't matter.

Safe examples:

Sum

$$(2+3)+4 = 2+(3+4)$$

Max

$\max(5,9,2)=9$

Min

Count

13. When NOT Safe (Image 4)

Average Problem

Numbers:

1,3,5

3,7,9

4,7,8,7

Local averages:

3

6.33

6.5

Then average of averages:

$(3 + 6.33 + 6.5)/3 = \text{wrong}$

Because group sizes are different.

14. Correct Average Method

Instead of sending average, combiner should send:

(sum, count)

Example:

(9,3)

(19,3)

(26,4)

Reducer computes:

$(9+19+26)/(3+3+4)$

$=54/10$

$=5.4$

Correct answer.

15. Interview Definition

Combiner in Hadoop

Combiner is an optional local aggregation step that runs after mapper to reduce data transferred to reducer.

16. Internal Flow

Input File



Split into Blocks



Mapper



Combiner (optional local reduce)



Shuffle + Sort



Reducer



Final Output

17. Key Interview Question

Is Combiner same as Reducer?

Often same code can be reused.

But logically:

Combiner	Reducer
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Local machine	Final aggregation
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Optional	Mandatory if reducer exists
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Optimization	Final result
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18. One-Line Memory Trick

Mapper creates data
Combiner shrinks data
Reducer finalizes data

19. What Your Images Were Teaching

They were showing:

- ✓ Why reducer becomes bottleneck
 - ✓ How combiner reduces reducer load
 - ✓ Why max/min/sum are safe
 - ✓ Why average can fail
 - ✓ Combiner = performance booster
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20. Final Industry Truth

In large Hadoop jobs:

Using combiner properly can reduce shuffle traffic by **70%+** sometimes.

That means:

- cheaper cluster cost
 - faster jobs
 - less reducer pressure
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21. Interview Smart Answer

If interviewer asks:

Why use Combiner?

Say:

To reduce intermediate data sent from mapper to reducer by doing local aggregation, improving MapReduce performance.